Unit 1 - Math Fundamentals
1.3 Functions

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MSDS 520: Math and Stats Foundations for Data Science
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Outline

1. Why Functions?
2. From Relations to Functions
3. Domain and Range
4. Families of Functions
5. Combining Functions
6. A Tool from Logic: Quantifiers
7. Inverse Functions
8. Functions of Several Variables
9. Functions in Machine Learning
10. Review

Why Functions?

A Problem to Motivate Us

Mixing an IV bag

"We need a 500 mL bag of saline at a concentration of 4 mg/mL. How many mg of drug go in?"

The rule is one you already know, mass is concentration times volume:

$$m \;=\; c \cdot V$$

Fix the bag at $V = 500$ mL and that rule becomes a machine: feed it a target concentration, it returns the mass to weigh out.

$$m(c) \;=\; 500\,c$$

That machine is a **function**. Running it forward ($c = 4 \Rightarrow m = 2000$) is **evaluation**; running it backward ("the vial holds 1500 mg: what concentration will that give?") is the **inverse**. Both are in this lecture.

Where This Shows Up in Data Science

Almost everything you will build is a function

- **A model is a function:** it takes a patient's measurements and returns a prediction.
- **Every data transformation is a function:** rescaling a column, taking a log, converting units.
- **Error is a function:** "how wrong is this model?" is a number that depends on the model's settings.
- **Training is a search over functions:** pick a family, then pick the best member of it.
- **Reading a result backward**: "what input would have given me this output?", is an inverse.

Unit 1-2 built **sets**. A function is what connects two sets: it sends each element of one to an element of the other.

What We Will Build

The plan

1. **Relations**: and what makes one a **function**.
2. **Domain** and **range**: which inputs are valid, which outputs occur.
3. The **families**: transformations, polynomials, roots, reciprocals, exponentials, logarithms, trigonometric and piecewise rules.
4. **Combining** functions, especially **composition**.
5. **Quantifiers**: the "for all" and "there exists" our definitions are written with.
6. **Inverses**: undoing a function, and when that is possible.
7. Functions of **several variables**: the **Cartesian product** they live on, and the **surfaces** they graph as.
8. Functions in **machine learning**: from a data table to a prediction.

Notebooks for this lecture: `U1-3_Functions-1_DomainRangeAndFamilies`, `U1-3_Functions-2_Composition`, `U1-3_Functions-3_Inverse`, `U1-3_Functions-4_FormulaExplorer`, `U1-3_Functions-5_SeveralVariables`.

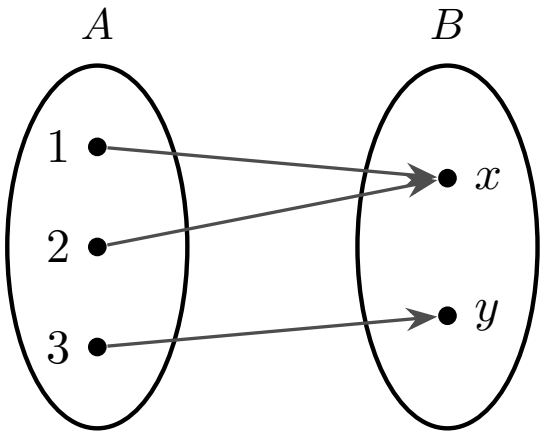
From Relations to Functions

Pairing Things Up

Unit 1-2 ended with **ordered pairs**: (a, b), where order matters, so $(1, 2) \neq (2, 1)$.

Relation

A **relation** R from a set A to a set B is any set of ordered pairs (a, b) with $a \in A$ and $b \in B$. If $(a, b) \in R$ we say "a is related to b."



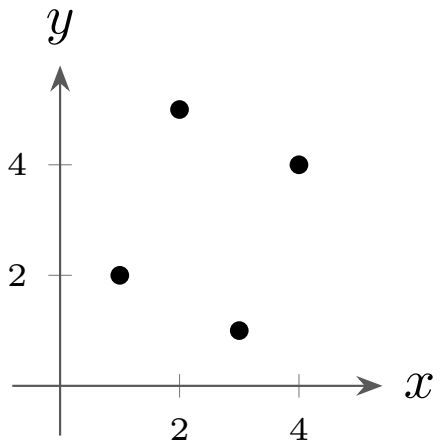
A relation is just **a set of links**. Drawing one arrow per pair turns the list into a picture:

$$R = \{(1, x),\ (2, x),\ (3, y)\}$$

Nothing here is required to be tidy, an element of A may link to many elements of B, to one, or to none at all.

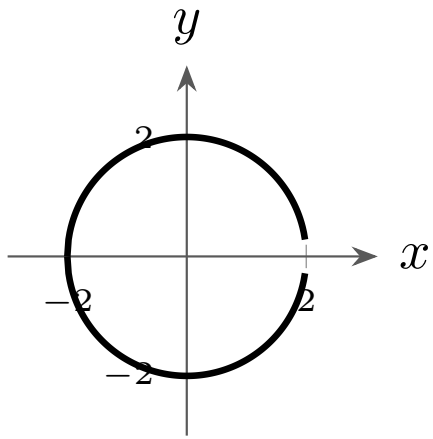
Relations Live in the Plane

When A and B are both $\mathbb{R}$, a pair (x, y) is a **point**, so a relation is a **picture**: every set of points you can draw is a relation.



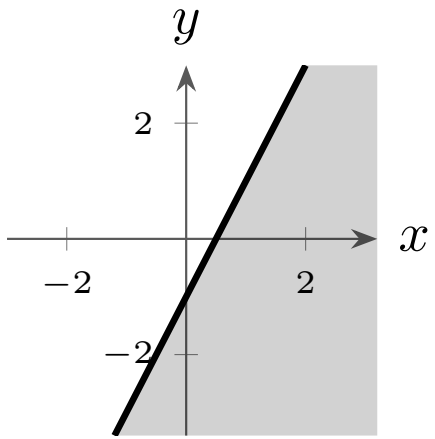
a handful of points

$R = \{(1, 2),\ (2, 5),\ (3, 1),\ (4, 4)\}$



a curve

$R = \{(x, y) \mid x^2 + y^2 = 4\}$



a whole region

$R = \{(x, y) \mid y \leq 2x - 1\}$

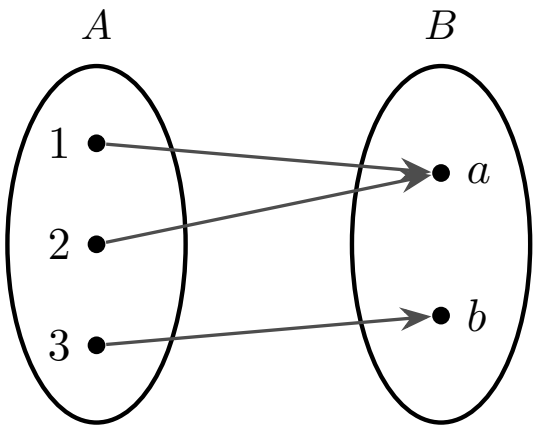
All three are perfectly good relations, a finite list, a curve, a shaded half-plane.

Only one of them will turn out to be a **function**.

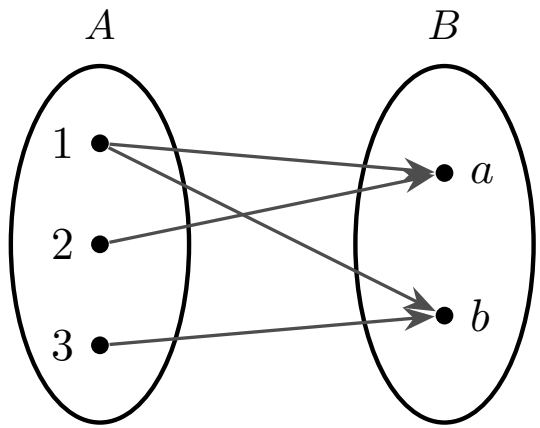
When Is a Relation a Function?

Function

A **function** f from A to B is a relation in which **every** element of A is related to **exactly one** element of B, that is, $\forall a \in A,\ \exists!\, b \in B$ with $(a, b) \in f$.



Is a function:
one arrow out of each input



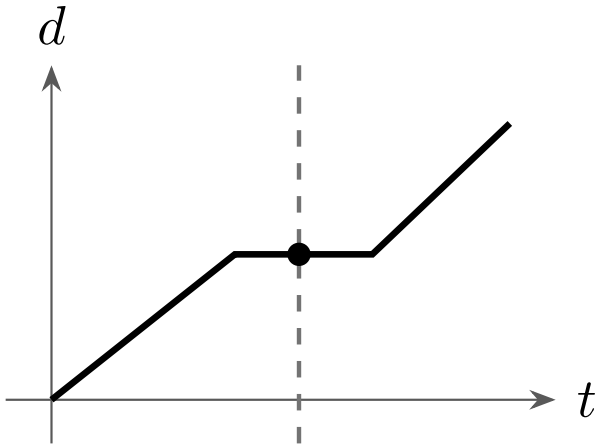
Not a function:
1 has two arrows out

The pictures are identical except for **one extra arrow**, $1 \to b$, that alone breaks it. Two inputs sharing an output is fine (**many-to-one**); one input with two outputs is not (**one-to-many**).

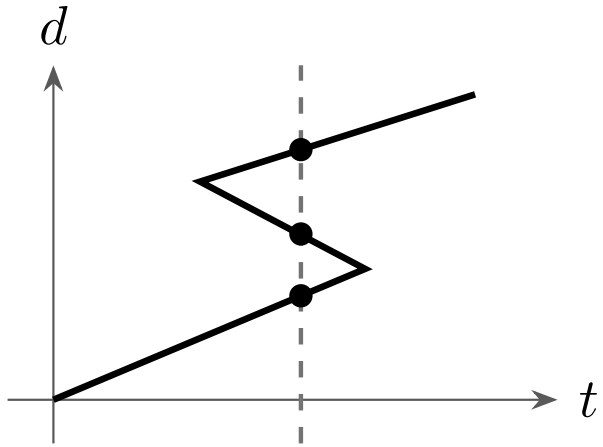
Why "Exactly One"? A Drive Across Town

You get in the car and drive. Let t be the **time** since you pulled out and $d(t)$ the **distance** you have covered by then. At any instant you are somewhere on the road, and somewhere **once**.

A real trip, *is* a function



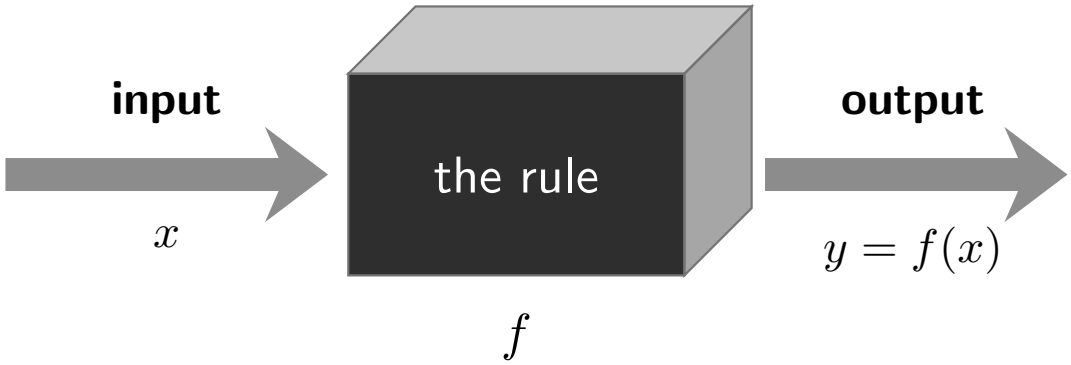
Two places at once, impossible



- The **flat stretch** is the red light you sat at: time ran on while d stayed put, so many times share one distance, **many-to-one**, legal.
- The right-hand graph doubles back: one instant meets **three** distances, so the car is in three places at once, **one-to-many**, not a function.

Repeats read **horizontally** are fine; repeats read **vertically** are not.

A Function Is a Machine



Feed in an x, the rule runs, one y comes out. Two demands hide inside that "exactly one":

- **At least one:** no input is left without an output, feed the machine a valid input and something must come out.
- **At most one:** no input produces two different outputs, feed it the same input twice and you get the same answer twice.

The IV bag rule $m(c) = 500\,c$ is exactly this: a box you drop a concentration into and read a mass out of. The rest of the lecture asks three questions about the box: what may go **in**, what comes **out**, and can we run it **backward**?

Notation

Reading $f : A \to B$

"f is a function from A to B." Here A is the **domain**, the valid inputs, and the outputs land in B.

If $f(x) = y$, then x is the **input** (or **argument**) and y is the **output**, the **value of f at x**.

A distinction worth keeping straight

f is the **rule**. $f(x)$ is the **number** that rule produces at x.
So "$f = x^2$" is sloppy; "$f(x) = x^2$" is right. Functions are usually named f, g, h, but any name works, and in code they get real names like `normalize` or `predict`.

The Same Function, Four Ways

Take $f(x) = 2x + 1$ on all of $\mathbb{R}$. The same function can be written down four ways:

As a formula	$f(x) = 2x + 1$	compact and exact
As a set of pairs	$\{\,(x,\, 2x+1) \mid x \in \mathbb{R}\,\}$	*every* pair, not a list
As a table	x | 0 1 2 $f(x)$ | 1 3 5	a **sample** of pairs
As a graph	the line through all the pairs	the whole picture

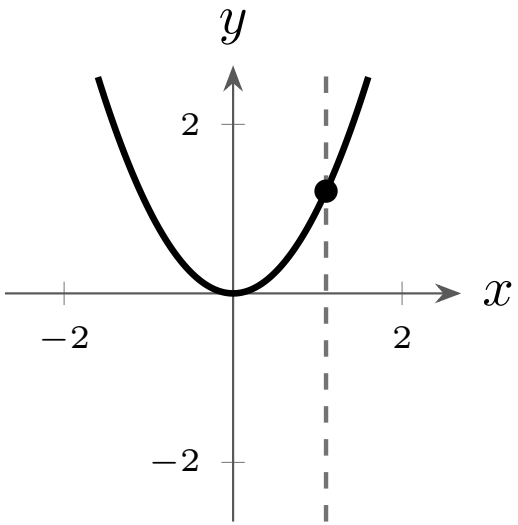
Only the table is incomplete. The domain here is all of $\mathbb{R}$, so the function really pairs *every* real with its output, including $f(0.5) = 2$ and $f(-\pi) = 1 - 2\pi$. A table can only ever show finitely many rows, which is why we still write the formula.

That gap is the job: real data arrives as a **table**, and modeling is guessing the formula behind it.

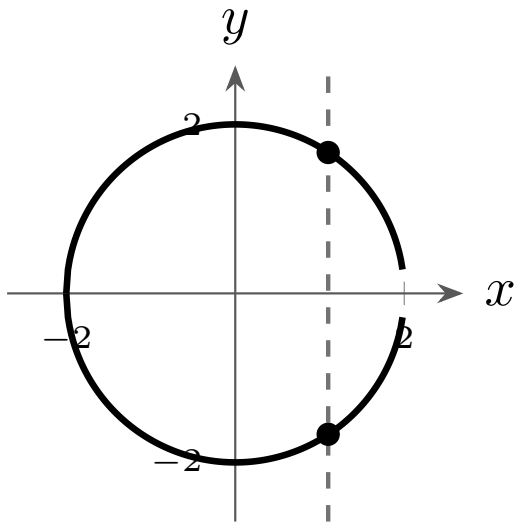
The Vertical Line Test

The graph of f is a picture of the pairs $(x, f(x))$. So "one output per input" has a visual meaning: **no vertical line may cross the graph twice**.

$y = x^2$, a function



$x^2 + y^2 = 4$, not a function



The circle fails: that one input meets **two** outputs, so "y as a function of x" is not defined here. The circle is a perfectly good *relation*, just not a function. Same reason a position-vs-time graph may never run vertically: one input, one output.

Exercise: Function or Not?

Try it: which of these are functions from A to B?

1. $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (2, 4), (3, 5)\}$
2. $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (1, 5), (2, 4), (3, 5)\}$
3. $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (3, 5)\}$
4. Each patient $\mapsto$ that patient's date of birth.
5. Each date of birth $\mapsto$ the patient born on it.
6. $y^2 = x$, taking x as the input and y as the output.

For each one that fails, say *which* of the two demands it breaks.

Exercise: Solution

Answers

1. **Yes.** All three inputs are used, one arrow each. (4 being hit twice is fine, many-to-one.)
2. **No.** 1 is paired with both 4 and 5: "at most one" fails.
3. **No.** 2 has no output at all: "at least one" fails.
4. **Yes.** A person has exactly one date of birth.
5. **No.** Many patients share a birthday, so one date leads to many patients, one-to-many.
6. **No.** The input $x = 4$ gives $y = 2$ **and** $y = -2$. This is the sideways parabola, and it fails the vertical line test.

Compare #4 and #5: reversing a function's arrows does **not** always give a function back. That single observation is the whole story of **inverses**, later in this lecture.

Domain and Range

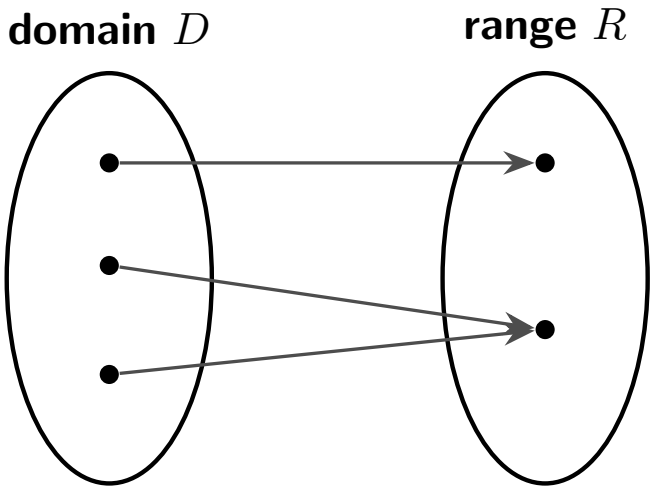
Domain and Range

Definitions, and the letters we will use

For a function f,

- the **domain** D is the set of every input f accepts;
- the **range** $R = \{\, f(x) \mid x \in D \,\}$, the outputs that actually **occur**.

When more than one function is in play, tag the letter: D_f, R_f, D_g, R_g.



every input
has an arrow out;
outputs may be shared

D asks *what may go in*, R asks *what comes out*, and the second is the harder question.

D Is Chosen, R Is Discovered

Example: $g(x) = x^2$

g accepts every real number, so $D = \mathbb{R}$. But no square is negative, so $R = [0, \infty)$, the entire negative half of the number line is never produced.

- D is **chosen**. It is part of saying what the function *is*, declaring g on all of $\mathbb{R}$, or only on $x \geq 0$, gives two different functions from the same formula.
- R is **discovered**. Once the rule and the domain are fixed, the range is whatever comes out; you do not get to pick it.

Which is why they are found differently

A domain is usually read off the **formula**: what would break? A range is usually read off the **graph**: how high does it go, how low, and does it skip anything in between?

What Restricts a Domain: the Mathematics

Often D is all of $\mathbb{R}$. Three operations spoil that:

The three usual suspects

1. **Division:** $\dfrac{1}{x}$ requires $x \neq 0$, no number times 0 gives 1. Exclude *every* value that kills the denominator: $\dfrac{1}{x^2-4}$ loses two, $x = \pm 2$.
2. **Even roots:** $\sqrt{x}$, $\sqrt[4]{x}$, $\sqrt[6]{x}$, ... require $x \geq 0$: an **even** power of a real number is never negative, so nothing squares to a negative. Note $\geq$, not $>$, $\sqrt{0} = 0$ is legal.
3. **Logarithms:** $\log(x)$ requires $x > 0$, **strictly**. The log asks "what exponent gives x?", and b^y is always positive, so no exponent yields 0 or a negative.

Odd roots are not a problem

$\sqrt[3]{x}$, $\sqrt[5]{x}$, ... are defined for **every** real number: $\sqrt[3]{-8} = -2$, since $(-2)^3 = -8$. An odd power keeps the sign of its base, so an odd root can undo a negative. **Only even indices restrict** D, $\sqrt[3]{x}$ has domain $\mathbb{R}$, no conditions at all.

What Restricts a Domain: the Context

Those rules give the **mathematical** domain, where the algebra is legal. The **practical** domain is usually smaller, because reality has rules the algebra never heard of.

Where the extra restrictions come from

- **Physical impossibility.** A dose or a concentration cannot be negative; if t is hours since admission then $t \geq 0$. The formula will still evaluate at $t = -3$ and report on a time that never happened.
- **Discreteness.** A count of patients is a whole number: $f(2.7)$ computes fine and means nothing.
- **Plausible range.** An age of 150 or a systolic of 400 is a data-entry error, not an input. A domain doubles as a **validation rule**: most pipeline "bugs" are values that should never have been let in.
- **Units.** $\mathrm{BMI}(w, h) = w/h^2$ expects kilograms and meters; feed it pounds and inches and every step is valid arithmetic on a meaningless answer.

A practical domain is part of what the function *is*, and a model's honest limit.

Finding a Domain: The Recipe

Procedure

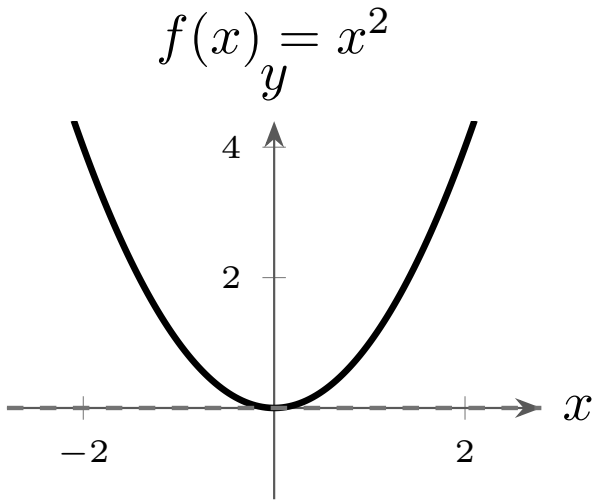
1. Find every division, even root, and logarithm in the formula.
2. Write the condition each one forces: denominator $\neq 0$, radicand ≥ 0, log argument > 0.
3. Solve each condition for x.
4. **Intersect** the results, all conditions must hold at once.
5. Write D in interval notation.

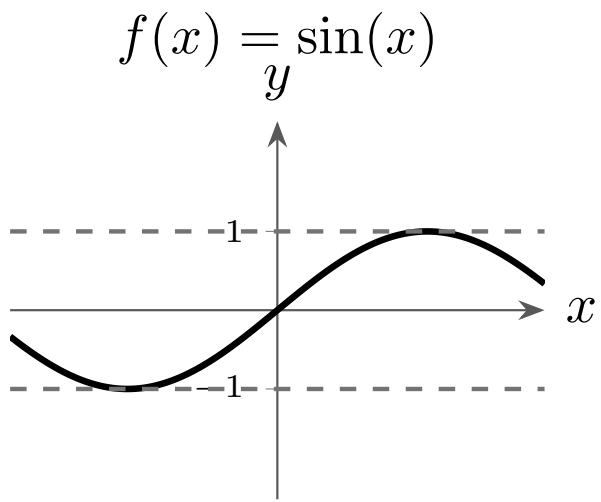
Step 4 is where Unit 1-2 pays off: "all conditions at once" is the **AND** of Unit 1-1, which is the **intersection** $\cap$ of the sets of allowed inputs.

Every **family** later in this lecture closes with worked examples of the whole recipe, domain *and* range, once its shape is on the board to read the range off of.

Range Is the Harder Half

D comes from the formula. R usually comes from **looking at the graph** and asking: how high does it go, how low, and does it skip anything in between?



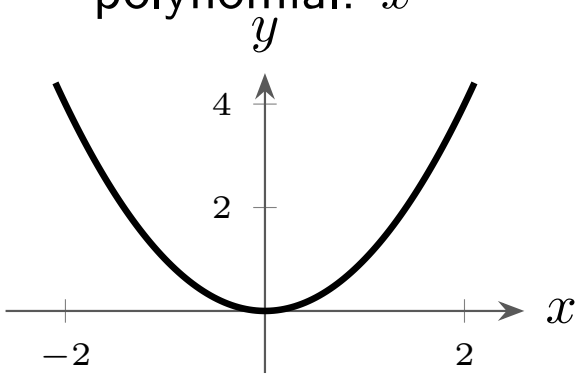


Left: the parabola bottoms out at 0 and rises forever, so $R = [0, \infty)$. Right: the sine wave is trapped between the two dashed lines, so $R = [-1, 1]$, a **bounded** function.

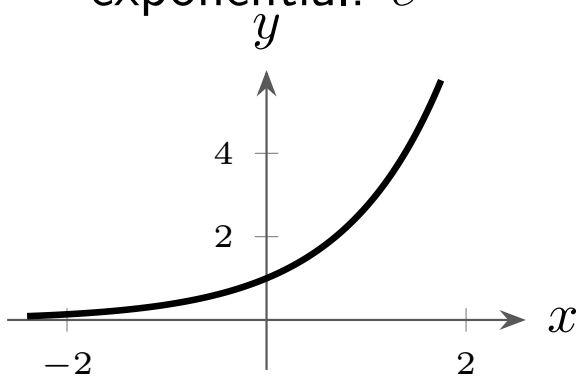
Families of Functions

Four Shapes Worth Memorizing

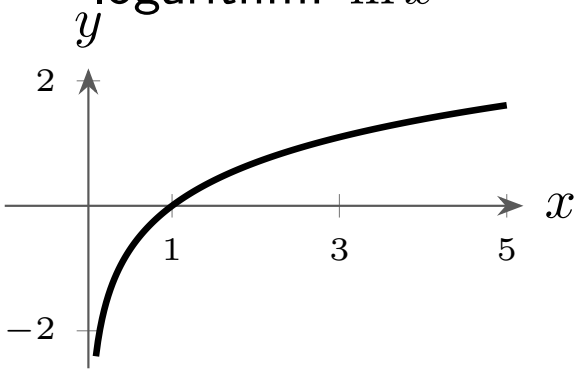
polynomial: x^2



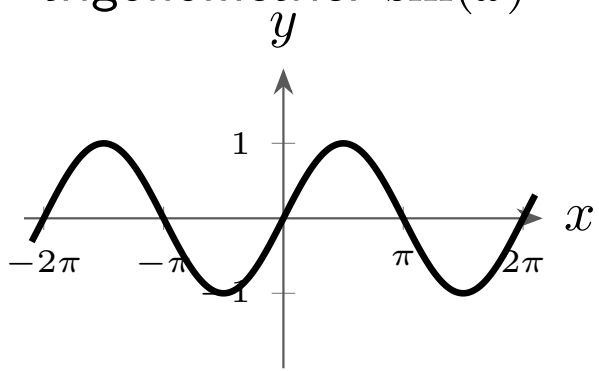
exponential: e^x



logarithm: $\ln x$



trigonometric: $\sin(x)$



Curves up in a bowl; shoots up; rises slowly and only for $x > 0$; repeats forever.

Recognizing these four at a glance is the goal, real data keeps producing them.

Transformations: One Family, Many Members

Starting from any $f(x)$, five small edits move and reshape its graph. Take $f(x) = x^2$ throughout:

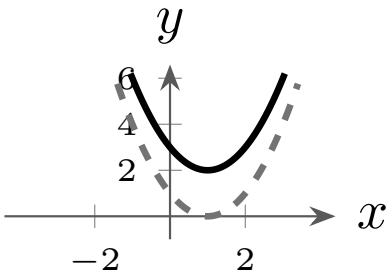
Edit	**Effect**	**Example**	**Watch out**
$f(x) + c$	shift **up** by c	$x^2 + 2$	vertical, as expected
$f(x + c)$	shift **left** by c	$(x + 2)^2$	**opposite** to the sign
$a\, f(x)$	stretch **vertically** by a	$3x^2$	taller if $a > 1$
$f(bx)$	squeeze **horizontally** by b	$(2x)^2$	narrower if $b > 1$
$f(-x)$	reflect across the y-axis	$(-x)^2 = x^2$	unchanged here!

Rows two and four catch people out for the same reason: an edit *inside* the function acts on the input, so it runs backwards. $f(x + 2)$ moves the graph **left**, and $f(2x)$ makes it **narrower**, since the input reaches any given value twice as fast.

Vertical edits do what they say; horizontal edits do the opposite. The last row is a third surprise: a parabola is **symmetric**, so flipping it left-right changes nothing.

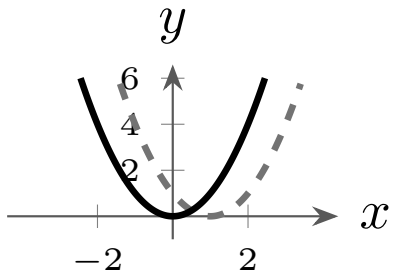
The Five Edits, Applied to $f(x) = (x-1)^2$

$f(x) + 2 = (x-1)^2 + 2$



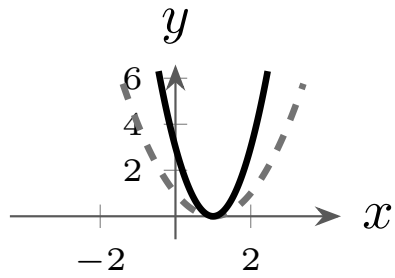
shift **up** 2

$f(x+1) = x^2$



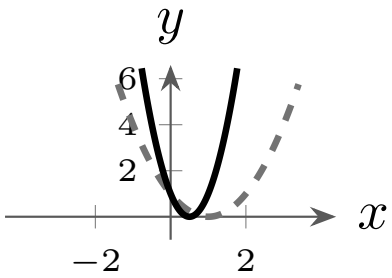
shift **left** 1

$3f(x) = 3(x-1)^2$



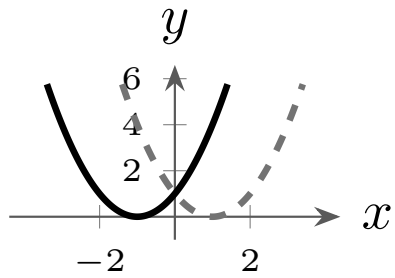
stretch $\times 3$ vertically

$f(2x) = (2x-1)^2$



squeeze $\times 2$ horizontally

$f(-x) = (x+1)^2$



reflect across y-axis

Dashed is $f(x) = (x-1)^2$ in every panel, solid is the edited version. The base is a *shifted* parabola so the horizontal edits have something to show: plain x^2 is symmetric.

Exercise: Transformations

Try it

1. Describe how the graph of $(x-3)^2+1$ differs from that of x^2.
2. Starting from $f(x)=\sqrt{x}$, write the formula that shifts the graph **left** 2 and then stretches it vertically by 3. What happens to D?
3. Which of the five edits leave the range unchanged, no matter what the function is?

Exercise: Solution

1. The shifted parabola

$(x-3)^2+1$ is the parabola moved **right** 3 and **up** 1, so its lowest point sits at $(3,1)$ instead of the origin. Right, not left, because the -3 is inside the function and horizontal shifts run opposite to their sign.

2. Building a formula

Shifting left 2 replaces x by $x+2$, then the stretch multiplies the whole thing: $3\sqrt{x+2}$. The root still needs a non-negative argument, so $x+2 \geq 0$ and $D=[-2,\infty)$: the horizontal shift dragged the domain along with it, while the stretch left it alone.

3. The edits that spare the range

The three **horizontal** edits: a shift $f(x+c)$, a squeeze $f(bx)$, and a reflection $f(-x)$ each feed the function the same values in a different order, so the set of outputs is untouched. The two vertical edits do change R: a shift slides it, a stretch scales it.

Linear Functions

Definition

$$f(x) = mx + b$$

m is the **slope** (rise over run, how much f changes per unit of x) and b is the **intercept** (the value at $x = 0$).

- Domain $\mathbb{R}$; range $\mathbb{R}$ when $m \neq 0$, and just $\{b\}$ when $m = 0$ (a flat line).
- **Constant rate of change:** every extra unit of x adds the same m to the output.
- This is the entire content of **linear regression**: choose m and b so the line sits as close as possible to the data.

Slope carries units and therefore meaning: "blood pressure rises 0.4 mmHg per additional kg" *is* a slope.

Polynomials

Definition

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{i=0}^{n} a_i x^i$$

The largest exponent n is the **degree**; the numbers a_i are the **coefficients**.

Degree	Name	Shape	Example
0	constant	flat line	$f(x) = 4$
1	linear	straight line	$f(x) = 2x - 1$
2	quadratic	one bowl (parabola)	$f(x) = x^2 - 3x + 2$
3	cubic	one S-bend	$f(x) = x^3 - x$

Domain is always $\mathbb{R}$, and a degree-n polynomial bends at most $n - 1$ times, so a higher degree chases the data more closely, and its *noise* with it.

Exercise: Polynomials

Try it

1. Give D and R for $g(x) = 3x^2 - 1$.
2. State the degree of $p(x) = 2x^3 - 5x + 1$, and say at most how many bends its graph can have.
3. Write p in summation notation, naming each coefficient a_i.

For #1, remember that a domain comes from the formula but a range comes from the shape.

Exercise: Solution

1. Domain and range of $3x^2 - 1$

A polynomial only adds and multiplies, so nothing is ever undefined: $D = \mathbb{R}$. The parabola opens upward and bottoms out where $x = 0$, giving the value -1, so $R = [-1, \infty)$.

2. Degree and bends

The largest exponent is 3, so p is a **cubic**. A degree-n polynomial bends at most $n - 1$ times, so p has at most **two** bends.

3. In summation notation

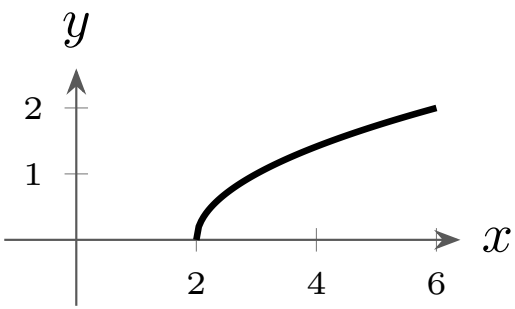
$$p(x) = \sum_{i=0}^{3} a_i x^i, \qquad a_0 = 1, \;\; a_1 = -5, \;\; a_2 = 0, \;\; a_3 = 2$$

The missing x^2 term is not absent, it is present with coefficient 0.

Roots and Reciprocals

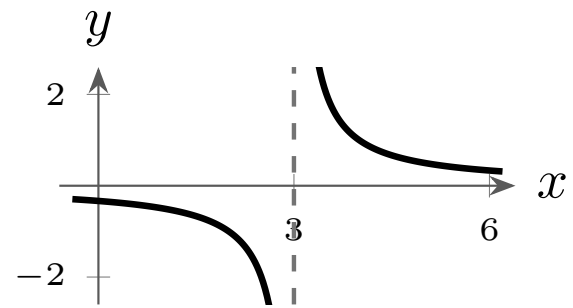
Not polynomials, but built from them, and they are the two shapes that **restrict** D.

$$f(x) = \sqrt{x-2}$$



$$D = [2, \infty), \quad R = [0, \infty)$$

$$g(x) = \frac{1}{x-3}$$



$$D = \mathbb{R} \setminus \{3\}, \quad R = \mathbb{R} \setminus \{0\}$$

Root: the recipe forces $x - 2 \geq 0$, and since a square root is never negative while $x - 2$ sweeps all of $[0, \infty)$, every non-negative output occurs.

Reciprocal: $x = 3$ is excluded, and the graph tears there (the dashed **asymptote**); the output gets as close to 0 as you like but never equals it.

Exercise: Roots and Reciprocals

Try it: give D in interval notation

1. $f(x) = \sqrt{5 - x}$ (then give R as well)
2. $g(x) = \dfrac{1}{x^2 - 4}$
3. $k(x) = \dfrac{\sqrt{x}}{x - 1}$ (careful, this one has **two** conditions)

Run the recipe: spot the roots and divisions, write each condition, solve, then **intersect**.

Exercise: Solution

Domains

1. $5 - x \geq 0 \Rightarrow x \leq 5$: $\quad D = (-\infty, 5]$.
2. $x^2 - 4 \neq 0 \Rightarrow x \neq \pm 2$: $\quad D = (-\infty, -2) \cup (-2, 2) \cup (2, \infty)$.
3. The root needs $x \geq 0$; the denominator needs $x \neq 1$. Intersecting: $D = [0, 1) \cup (1, \infty)$.

Range of #1

A square root is never negative, and $5 - x$ covers all of $[0, \infty)$ as x runs down from 5, so every non-negative value occurs: $\quad R = [0, \infty)$.

#2 is worth a second look: a **squared** expression in the denominator excludes *two* points, not one.

Exponential Functions

Definition

$$f(x) = a\,b^{x}, \qquad b > 0,\ b \neq 1$$

a is the starting value at $x = 0$, and b is the **growth factor** per unit step.

- $b > 1$: **growth**, each step **multiplies** by b. $b < 1$: **decay**.
- $D = \mathbb{R}$; $R = (0, \infty)$ for $a > 0$, an exponential never reaches 0.
- The natural base is $e \approx 2.718$, and e^{x} is the default choice throughout math and machine learning.

Linear vs. exponential, in one sentence

A linear function **adds** the same amount each step; an exponential **multiplies** by the same factor each step.

Drug clearance, bacterial growth, and compound interest are all exponential.

Exercise: Exponentials

Try it

1. Give D and R for $f(x) = 3^x$.
2. A colony starts at 200 cells and doubles every hour. Write $N(t)$, the count after t hours.
3. Is $c(x) = 3^{-x}$ growth or decay? Give its range.

Ask yourself for #3 whether the output can ever be zero or negative.

Exercise: Solution

1. Domain and range of 3^x

Any real exponent is allowed, so $D = \mathbb{R}$. A positive base raised to any power stays positive, and every positive value is achieved somewhere, so $R = (0, \infty)$: the curve approaches the x-axis but never touches it.

2. The colony

Doubling each hour means **multiplying** by 2 each step, so the model is exponential with starting value 200 and growth factor 2:

$$N(t) = 200 \cdot 2^t \qquad (N(0) = 200,\ N(1) = 400, \text{ as required})$$

3. Growth or decay

$3^{-x} = (1/3)^x$, a growth factor below 1, so it is **decay**. The range is unchanged: $R = (0, \infty)$. Flipping the sign of the exponent reflects the curve left to right; it does not let the output reach 0.

Logarithmic Functions

Definition, the log answers "what exponent?"

$$\log_b(y) = x \qquad \text{means exactly} \qquad b^x = y$$

So $\log_2(8) = 3$, because $2^3 = 8$. The logarithm **undoes** the exponential, our first real example of an inverse.

- $D = (0, \infty)$, $R = \mathbb{R}$, the exponential's, swapped.
- Common bases: $b = e$ (written $\ln$), $b = 10$, $b = 2$ (Unit 1-1's information measures).
- The useful rules: $\log(uv) = \log u + \log v, \quad \log(u^p) = p \log u.$

Worked example: $h(x) = \ln(x + 1)$

The argument must be positive: $x + 1 > 0$, so $D = (-1, \infty)$. The log still climbs and falls without bound, so $R = \mathbb{R}$, shifting the input moves D, not R.

The first rule is why we **log-transform** skewed data.

Exercise: Logarithms

Try it

1. Give D and R for $h(x) = \log_{10}(x)$.
2. Give D and R for $\ell(x) = \ln(3 - x)$.
3. Solve $\log_2(x) = 5$, and say which function you used to undo the log.

For #2, the recipe applies to whatever sits *inside* the logarithm, not to x itself.

Exercise: Solution

1. $\log_{10}(x)$

The argument must be strictly positive, so $D = (0, \infty)$. The log climbs and falls without bound, so $R = \mathbb{R}$: exactly the exponential's domain and range, swapped.

2. $\ln(3 - x)$

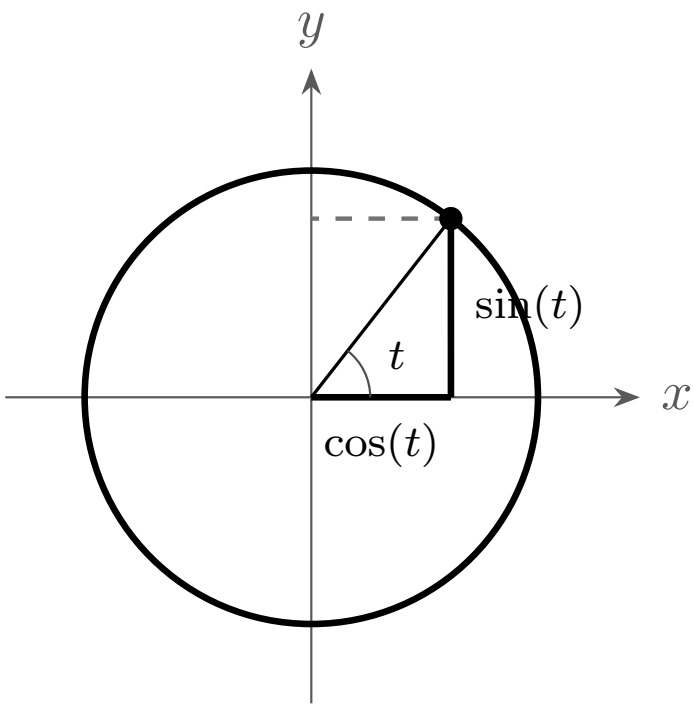
Now the *argument* must be positive: $3 - x > 0$, so $x < 3$ and $D = (-\infty, 3)$. As x climbs toward 3 the argument shrinks toward 0 and the log dives; as x runs off to the left the argument grows without bound. So $R = \mathbb{R}$ still: shifting and flipping the input moves D, not R.

3. Solving $\log_2(x) = 5$

The exponential undoes the log: raise 2 to both sides, giving $x = 2^5 = 32$.
Check: $\log_2(32) = 5$ because $2^5 = 32$.

Trigonometric Functions: the Unit Circle

Sine and cosine come from one picture: walk a point counter-clockwise around the **unit circle** (radius 1, centre at the origin), and let t be the angle turned.



Pythagoras on those two legs gives $\cos^2(t) + \sin^2(t) = 1$ for free.

Definition

For the point at angle t on the unit circle,

- $\cos(t)$ is its x-**coordinate**,
- $\sin(t)$ is its **height**, the y-coordinate.

Two facts fall straight out of the picture:

- Going once around is 2π and lands you where you started, so both functions are **periodic**: $\sin(t + 2\pi) = \sin(t)$.
- The point never leaves the circle, so neither coordinate can exceed 1: both are **bounded**, with $R = [-1, 1]$.

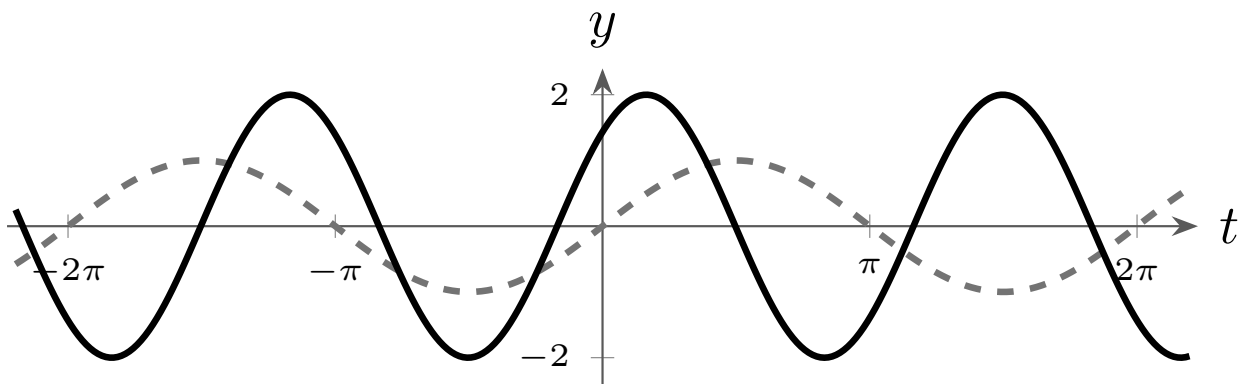
The General Wave

Plotting the height $\sin(t)$ against the angle t unrolls the circle into a wave. Three constants bend that wave into any shape you need:

Three knobs

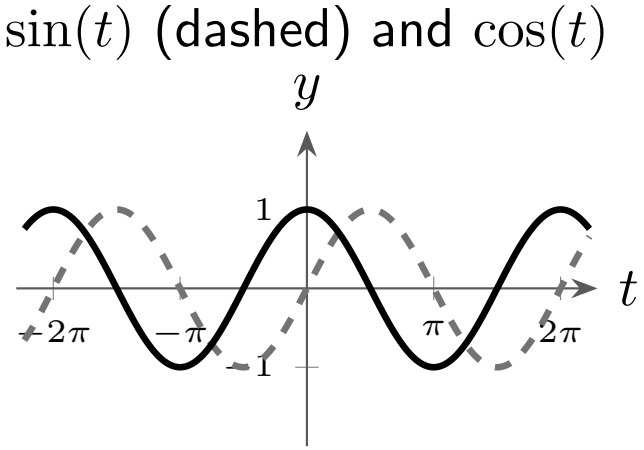
$$f(t) = A\sin(\omega t + \phi)$$

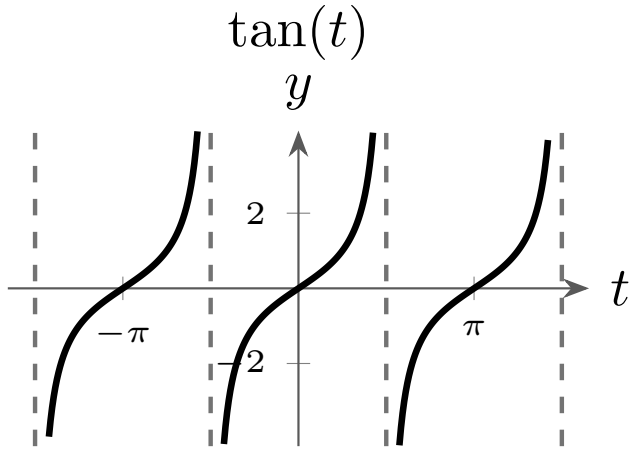
A is the **amplitude** (how tall), ω the **angular frequency** (how fast it cycles, the period is $2\pi/\omega$), and ϕ the **phase**: where in the cycle it starts, sliding the wave left by ϕ/ω.



Dashed is plain $\sin(t)$, solid has all three knobs turned; $D = \mathbb{R}$ and $R = [-A, A]$. Anything cyclic is modeled this way, and **fitting** one means choosing A, ω, ϕ.

Cosine and Tangent





- **Cosine** is sine with a phase of $\pi/2$: $\cos(t) = \sin\left(t + \frac{\pi}{2}\right)$. Same shape, same $D = \mathbb{R}$ and $R = [-1, 1]$; it simply starts at its peak instead of at zero.
- **Tangent** is their ratio, $\tan(t) = \dfrac{\sin(t)}{\cos(t)}$, a **division**, so the domain rules from earlier apply: it is undefined wherever $\cos(t) = 0$, that is at $t = \frac{\pi}{2} + k\pi$ for every integer k (dashed lines).
- So $D_{\tan} = \{t \in \mathbb{R} : t \neq \frac{\pi}{2} + k\pi\}$ and $R_{\tan} = \mathbb{R}$, unlike sine and cosine, tangent is **not bounded**, and its period is π, not 2π.

Exercise: Trigonometric Functions

Try it, with $k(t) = 4\sin(2t + \pi)$

1. Name the amplitude, the angular frequency, and the phase, then give the period.
2. Give D and R for k.
3. Why is $\tan(t)$ undefined at $t = \pi/2$, and what does its graph do there?
4. On the unit circle, what are $\cos(0)$ and $\sin(0)$? Read them off the picture rather than a calculator.

Exercise: Solution

1 and 2. Reading the wave

Matching $A\sin(\omega t + \phi)$ term by term: amplitude $A = 4$, angular frequency $\omega = 2$, phase $\phi = \pi$. The period is $2\pi/\omega = \pi$, so the wave completes a full cycle twice as fast as $\sin(t)$.

Every real t is a legal angle, so $D = \mathbb{R}$, and the sine is trapped between ± 1 before the amplitude scales it, so $R = [-4, 4]$.

3 and 4. Tangent, and two values off the circle

$\tan(t) = \sin(t)/\cos(t)$ is a division, and $\cos(\pi/2) = 0$, so the rule asks for a division by zero: $t = \pi/2$ is excluded from D. The graph tears there, running off toward $+\infty$ on one side and $-\infty$ on the other, which is the **asymptote**.

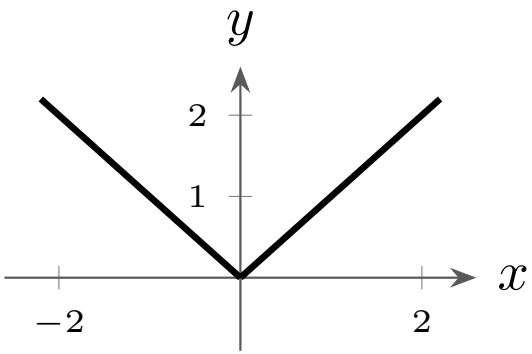
At angle 0 the point sits at the far right of the circle, at coordinates $(1, 0)$: its x-coordinate gives $\cos(0) = 1$ and its height gives $\sin(0) = 0$.

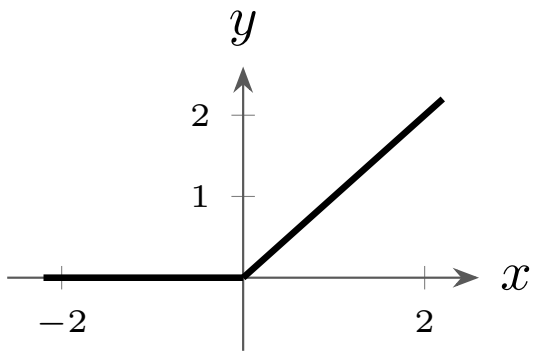
Piecewise Functions

A function may be defined by **different rules on different parts of its domain**, so long as each input still gets exactly one output. Read the conditions on the right first: they say *which* rule applies.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\mathrm{ReLU}(x) = \begin{cases} x, & x > 0 \\ 0, & x \leq 0 \end{cases}$$



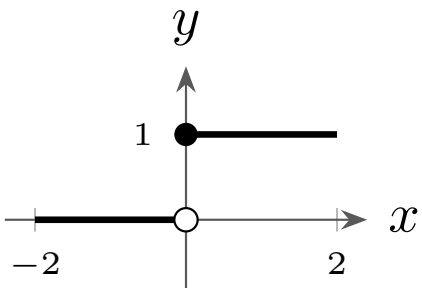


Both pieces meet, so these are **continuous**, but each has a **corner** at the origin. ReLU, "keep it if positive, otherwise zero," is the workhorse activation inside neural networks.

More Piecewise Examples

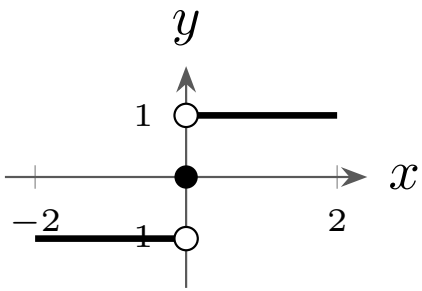
unit step (Heaviside)

$$H(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



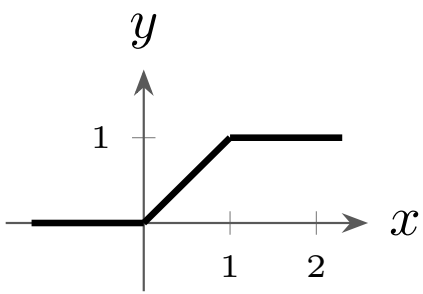
sign

$$\operatorname{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$



clipping (clamp)

$$c(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$



- The first two **jump** at $x = 0$: the pieces do not meet. The **filled** dot is the value the function takes there; the **hollow** dot is a value it does not take.
- Clipping doesn't jump, but it flattens. It is how you cap a dose, a pixel value, or a probability at its legal limits.

A threshold rule, "treat if the reading is above the cutoff", is a step function.

Exercise: Piecewise Functions

Try it

1. Give D and R for ReLU.
2. Give D and R for the clipping rule c, which returns 0 below 0, x on $[0, 1]$, and 1 above 1.
3. Give D and R for the unit step H. What is unusual about this range?
4. Using the five edits, describe the graph of $\mathrm{ReLU}(x - 2) + 1$ and give its D and R.

A piecewise rule is still one function, so it still has exactly one D and one R: collect the pieces, then take stock of everything they produce together.

Exercise: Solution

1 and 2. Two continuous rules

ReLU: every real number gets an answer, so $D = \mathbb{R}$. Negative inputs return 0 and positive ones return themselves, so $R = [0, \infty)$.

Clipping: again $D = \mathbb{R}$, but the output is held between the two caps, so $R = [0, 1]$. This is the point of clipping: it *chooses* the range.

3. A range that is not an interval

$D = \mathbb{R}$, and the only values ever produced are 0 and 1, so $R = \{0, 1\}$: two isolated points. A range is a **set** of outputs, and nothing requires it to be an interval.

4. Moving a piecewise graph

$\mathrm{ReLU}(x - 2) + 1$ is the same bent line shifted **right** 2 and **up** 1, so its corner sits at $(2, 1)$. The domain is untouched, $D = \mathbb{R}$, while the vertical shift carries the range with it: $R = [1, \infty)$.

Combining Functions

The Easy Combinations

Given two functions f and g, we can build new ones point by point:

$$(f+g)(x) = f(x) + g(x), \qquad (fg)(x) = f(x)\,g(x), \qquad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Watch the domain

The new function is only defined where **both** pieces are, and division adds the condition $g(x) \neq 0$. So

$$D_{f+g} \;=\; D_f \cap D_g$$

another appearance of the intersection from Unit 1-2.

These are useful but not deep. The combination that really matters is the next one.

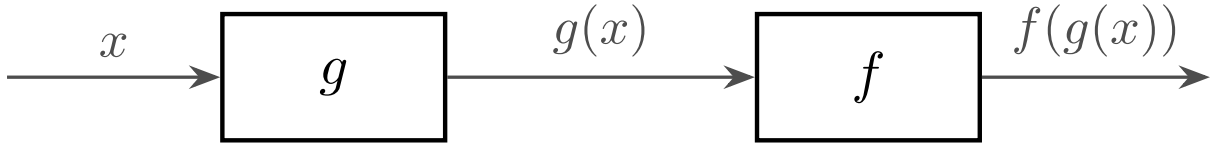
Composition: Feeding One Function into Another

Definition

The **composition** of f with g is

$$(f \circ g)(x) = f\big(g(x)\big)$$

read "f of g of x" or "f after g."



Work from the inside out. g runs first, even though f is written first. The parentheses tell you the order, not the reading direction.

A Worked Composition

Let $f(x) = x^2$ and $g(x) = x + 3$.

Evaluate $(f \circ g)(2)$ step by step

$$\begin{aligned}
(f \circ g)(2) &= f\big(g(2)\big) && \text{write it out} \\
&= f(2+3) && \text{inner function first} \\
&= f(5) && \\
&= 5^2 \;=\; 25 && \text{outer function second}
\end{aligned}$$

Now as a formula, for every x

$$(f \circ g)(x) = f(x+3) = (x+3)^2$$

Substitute the *whole* inner expression wherever the outer function had its input, parentheses included.

Order Matters

With the same $f(x) = x^2$ and $g(x) = x + 3$, swap the order:

	Formula	**At** $x = 2$
$(f \circ g)(x) = f(g(x))$	$(x+3)^2$	25
$(g \circ f)(x) = g(f(x))$	$x^2 + 3$	7

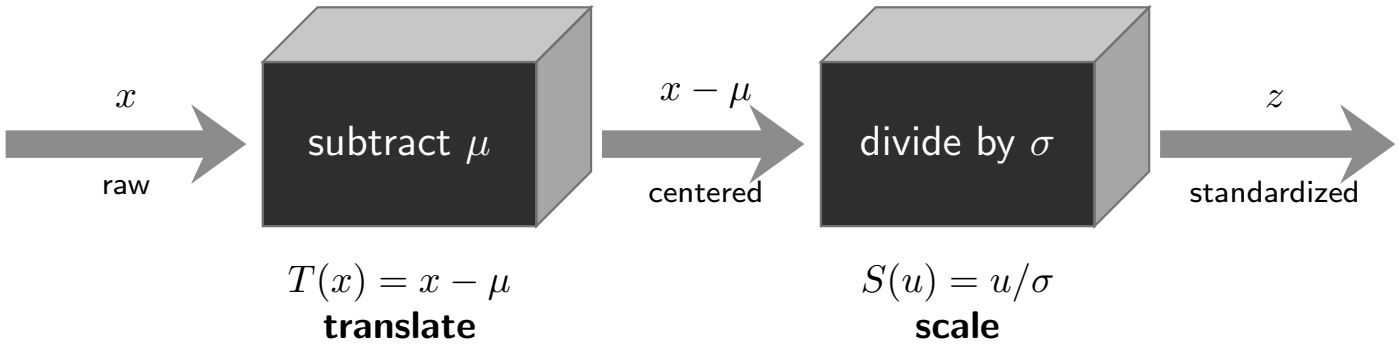
Composition is not commutative

In general $f \circ g \neq g \circ f$.

The plain-language version: "add 3, then square" is not the same instruction as "square, then add 3." Real pipelines have the same hazard, removing outliers and then rescaling gives a different result from rescaling and then removing outliers.

Chains of Steps: the z-Score

Composition is what a **pipeline** is. Standardizing a measurement, rewriting it as "how many standard deviations from the mean", is two machines in a row:



$$z \;=\; (S \circ T)(x) \;=\; S\big(T(x)\big) \;=\; \frac{x - \mu}{\sigma}$$

T for **translate** and S for **scale** are the usual names for these two moves. Translate first, then scale: after T the average input sits at 0; after S one unit of output means one standard deviation. To standardize **new** data, re-run both boxes in order with the *same* μ and σ.

Reading a Composition Backward

Just as useful as building compositions is **spotting** them: given a complicated function, name the inside and the outside. Write the outer rule in its own variable y, the value the inner function handed it.

Function	**Inner** $y = g(x)$	**Outer** $f(y)$
$\sqrt{x^2+1}$	x^2+1	$\sqrt{y}$
$\ln(3x)$	$3x$	$\ln y$
$\sin^2(x)$	$\sin(x)$	y^2
e^{-x^2}	$-x^2$	e^y

The domain of a composition needs both stages to be valid: for $\ln(3x)$ the inner value $3x$ must land in the log's domain, forcing $x > 0$.

Hold onto this skill. In Unit 3 the **chain rule** differentiates a composition by handling the inner and outer pieces separately, and it starts by asking exactly this question.

Exercise: Composition

The functions

$$f(x) = 2x - 1, \qquad g(x) = x^2, \qquad h(x) = \sqrt{x}$$

Try it

1. Compute $(f \circ g)(3)$ and $(g \circ f)(3)$.
2. Write formulas for $(f \circ g)(x)$ and $(g \circ f)(x)$.
3. Give a formula for $(h \circ g \circ f)(x)$, and state its domain D.
4. Identify an inner $y = g(x)$ and an outer $f(y)$ for $1/\sqrt{x-4}$.

Work #1 and #2 **inside out**: evaluate the inner function first, then feed its output to the outer one.

Exercise: Solution

1 and 2

$(f \circ g)(3) = f(9) = 17$, while $(g \circ f)(3) = g(5) = 25$. Different, as expected.
$(f \circ g)(x) = 2x^2 - 1$ and $(g \circ f)(x) = (2x-1)^2$.

3. A three-stage chain

Inside out: f first, then g, then h:

$$(h \circ g \circ f)(x) = h\big((2x-1)^2\big) = \sqrt{(2x-1)^2} = |2x-1|$$

Domain: $(2x-1)^2$ is never negative, so the root is always valid, the domain is $\mathbb{R}$. The absolute value appears because a square root returns the non-negative root.

4. Decomposition

Inner $y = x - 4$, then $\sqrt{y}$, then $1/y$. The domain needs $x - 4 > 0$, strictly, since that value also sits in a denominator. Domain: $(4, \infty)$.

A Tool from Logic: Quantifiers

Statements That Depend on a Variable

Unit 1-1 dealt with **propositions**, sentences that are definitely true or definitely false. But "$x > 3$" is neither, until you say something about x.

Predicate

A **predicate** $p(x)$ is a statement whose truth depends on the value of x. It becomes a proposition once x is pinned down.

There are two ways to pin it down. **Substitute** a value, $p(5)$ is true, $p(1)$ is false, or say **how many** values make it true. That second move is a **quantifier**, and there are only two of them:

Symbol	**Name**	**Read as**
$\forall$	universal	"for all", "for every"
$\exists$	existential	"there exists", "for some"

"*Every* input gets *exactly one* output" is two quantifiers in one breath, and **inverses**, next, is where writing that down pays off.

The Universal Quantifier: $\forall$

Notation

$\forall x \in S,\ p(x)$ means "$p(x)$ is true for **every** x in S."

- $\forall x \geq 0,\ \sqrt{x}$ is defined — true
- $\forall x \in \mathbb{R},\ x^2 \geq 0$ — true
- $\forall x \in \mathbb{R},\ \sqrt{x}$ is defined — **false**

Naming the set is not optional

The first and third statements differ only in *where x is allowed to live*, and that difference flips the truth value. $x = -4$ is the whole objection to the third.

One counterexample destroys a $\forall$. No matter how many cases work, a single failure makes the universal claim false, which is why "it worked on my test set" is not a proof of anything.

The Existential Quantifier: $\exists$

Notation

$\exists x \in S$ such that $p(x)$ means "**at least one** x in S makes $p(x)$ true."

- $\exists x \in \mathbb{R}$ such that $x^2 = 2$ true, take $x = \sqrt{2}$
- $\nexists x \in \mathbb{R}$ such that $x^2 = -1$ true, no real number squares to a negative

The struck-through $\nexists$ says "there does not exist." Note also the **asymmetry of effort**: exhibiting one example settles an $\exists$, while denying one means ruling out every candidate.

Exactly one: $\exists!$

$\exists!\, x \in S$ such that $p(x)$ means there is **one and only one** such x, it exists, and no second one does. This is the quantifier the definition of a function is built on.

Order Matters When Quantifiers Stack

Two quantifiers in a row are read **left to right**, and swapping them changes the meaning entirely:

The same symbols, two different claims

$\forall x \in \mathbb{R},\ \exists y \in \mathbb{R}$ such that $y > x$ **true**

For each x, pick $y = x + 1$. The y is allowed to depend on the x.

$\exists y \in \mathbb{R}$ such that $\forall x \in \mathbb{R},\ y > x$ **false**

This demands one fixed y that beats *every* real number at once. There is none.

The plain-English version

"Everyone has a mother" versus "someone is everyone's mother." Both are $\forall$ and $\exists$ about the same relationship, only the order differs, and only one is true.

The rule: a variable introduced *later* may depend on the ones before it, never the other way round.

Negating a Quantifier

Denying a quantified statement **flips the quantifier** and negates what is inside:

$$\neg\big(\forall x,\ p(x)\big) \quad \equiv \quad \exists x \text{ such that } \neg p(x)$$

$$\neg\big(\exists x \text{ such that } p(x)\big) \quad \equiv \quad \forall x,\ \neg p(x)$$

Read them in words

"Not every patient responded" says exactly "**some** patient did not respond", not "no patient responded," which is a far stronger claim and a common misreading.

"There is no cure" says "**every** candidate fails to cure."

Same shape as De Morgan's laws from Unit 1-1: push the negation inward and every connective turns into its partner. Here $\forall$ and $\exists$ are the partners.

This is also how you **disprove** a universal claim in practice, negating it turns the task into finding a single counterexample.

Exercise: Quantifiers

Try it

1. Write in symbols: "every real number has a cube root" and "some integer is even."
2. True or false: $\forall x \in \mathbb{R},\ x^2 > 0$. Justify your answer.
3. Negate: "$\forall$ patients p, the drug lowered p's blood pressure." What would you have to find to prove the original false?
4. One of these is true and one is false. Which is which?

$$\forall x \in \mathbb{R},\ \exists y \in \mathbb{R} \text{ s.t. } y = x^2 \qquad \exists y \in \mathbb{R} \text{ s.t. } \forall x \in \mathbb{R},\ y = x^2$$

Exercise: Solution

1 and 2

$\forall x \in \mathbb{R}$, $\exists y \in \mathbb{R}$ such that $y^3 = x$. and $\exists n \in \mathbb{Z}$ such that n is even.

False, by the single counterexample $x = 0$, where $x^2 = 0$ is not > 0. The claim becomes true if you either weaken the predicate to $x^2 \geq 0$ or shrink the set to $x \neq 0$, **the set and the predicate are both part of the statement**.

3 and 4

Negation: "$\exists$ a patient p such that the drug did **not** lower p's blood pressure." To disprove the original, produce **one** such patient, not a whole failing trial.

The first is **true**: every real number has a square, and y may depend on x. The second is **false**: it asks for one fixed y equal to x^2 for *every* x at once, which fails already at $x = 1$ and $x = 2$.

Why We Needed This

Every definition in this lecture has been a quantified sentence in disguise. Here they are written properly, the last two are the ones the next section runs on:

Three definitions, in symbols

- **Function** $f : A \to B$, every input, exactly one output:

$$\forall a \in A,\ \ \exists!\, b \in B \text{ such that } f(a) = b$$

- **One-to-one** (injective) , no output is used twice:

$$\forall x_1, x_2 \in A,\ \ f(x_1) = f(x_2) \implies x_1 = x_2$$

- **Onto** (surjective) , nothing in B is left unhit:

$$\forall b \in B,\ \ \exists a \in A \text{ such that } f(a) = b$$

Both at once is a **bijection**, and those are exactly the functions that can be **inverted**, which is the next section.

Inverse Functions

Undoing a Function

Temperature conversion runs both directions:

$$F(c) = \tfrac{9}{5}c + 32 \qquad \text{and back} \qquad C(f) = \tfrac{5}{9}(f - 32)$$

Run one and then the other and you are back where you started:

$$C\big(F(c)\big) = c \qquad \text{for every } c.$$

Inverse function

If $f : A \to B$, its **inverse** $f^{-1} : B \to A$ is the function that undoes it:

$$f^{-1}\big(f(x)\big) = x \qquad \text{for every } x \text{ in the domain,}$$
$$f\big(f^{-1}(y)\big) = y \qquad \text{for every } y \text{ in the range.}$$

An inverse reverses every arrow in the mapping diagram. The whole question is whether the result is still a function.

A Warning About the Notation

f^{-1} does not mean $1/f$

$$f^{-1}(x) \neq \frac{1}{f(x)}$$

The superscript -1 here means "the undoing function," not "raised to the power -1."

For $f(x) = 3x$:

- the **inverse** is $f^{-1}(x) = x/3$ (divide by three, undoing the multiplication);
- the **reciprocal** is $1/f(x) = 1/(3x)$, a completely different function.

Unfortunately $\sin^2(x)$ *does* mean $\big(\sin(x)\big)^2$, while $\sin^{-1}(x)$ means the inverse sine. The exponent -1 is the exception, and you simply have to remember it.

When Does an Inverse Exist?

Reversing the arrows can break the "exactly one output" rule, so not every function has an inverse.

One-to-one (injective)

f is **one-to-one** if different inputs always give different outputs:

$$\forall x_1, x_2 \in A, \quad x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

If two inputs shared an output, reversing would send that output to **two** places, one-to-many, not a function.

Nothing is left without an output

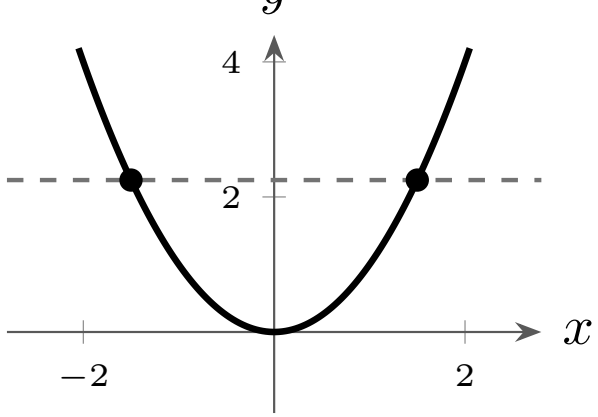
The other demand takes care of itself: f^{-1} only ever has to answer for values f **actually produces**, its **range**, and every one of those came from some input.

So **one-to-one is exactly the condition**: such an f has an inverse, and that inverse carries the range of f back to its domain.

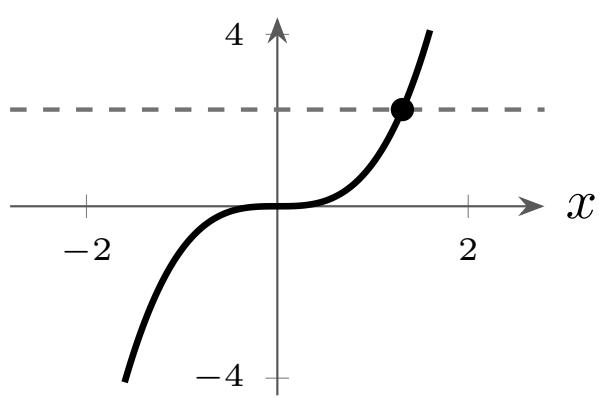
The Horizontal Line Test

One-to-one has a picture, just as "is a function" did: **no horizontal line may cross the graph twice**.

x^2, **not** one-to-one



x^3, one-to-one



Squaring loses the sign, so two inputs share each positive output and the square has no inverse on all of $\mathbb{R}$. Restrict the domain to $x \geq 0$ and it becomes one-to-one, which is why $\sqrt{\cdot}$ is defined as the **non-negative** root.

Finding an Inverse by Algebra

Procedure

1. Write $y = f(x)$.
2. **Solve for** x in terms of y.
3. Swap the names: the expression in y is $f^{-1}(y)$; rename y to x if you prefer.
4. Check: does $f^{-1}(f(x))$ simplify to x?

Example: $f(x) = 3x + 5$

$$
\begin{aligned}
y &= 3x + 5 \\
y - 5 &= 3x && \text{undo the } +5 \\
\tfrac{1}{3}(y - 5) &= x && \text{undo the } \times 3
\end{aligned}
$$

so $f^{-1}(x) = \dfrac{x-5}{3}$, and $f^{-1}(3x+5) = \dfrac{(3x+5)-5}{3} = x$. ✓

Undo in the Opposite Order

Notice what the algebra actually did. To build $f(x) = 3x + 5$ you **multiply by 3, then add 5**. To undo it you **subtract 5, then divide by 3**, the opposite operations, in the opposite order.

$$x \xrightarrow{\times 3} 3x \xrightarrow{+5} 3x + 5$$

$$\frac{y-5}{3} \xleftarrow{\div 3} y - 5 \xleftarrow{-5} y$$

Same idea as taking off a coat and then a sweater: reversing a composition reverses the order of its steps, $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$. It is also why a preprocessing pipeline has to be undone stage by stage, backward, to turn a prediction back into the original units.

Domain and Range Trade Places

Reversing every arrow swaps the roles of the two sets:

$$D_{f^{-1}} = R_f, \qquad R_{f^{-1}} = D_f$$

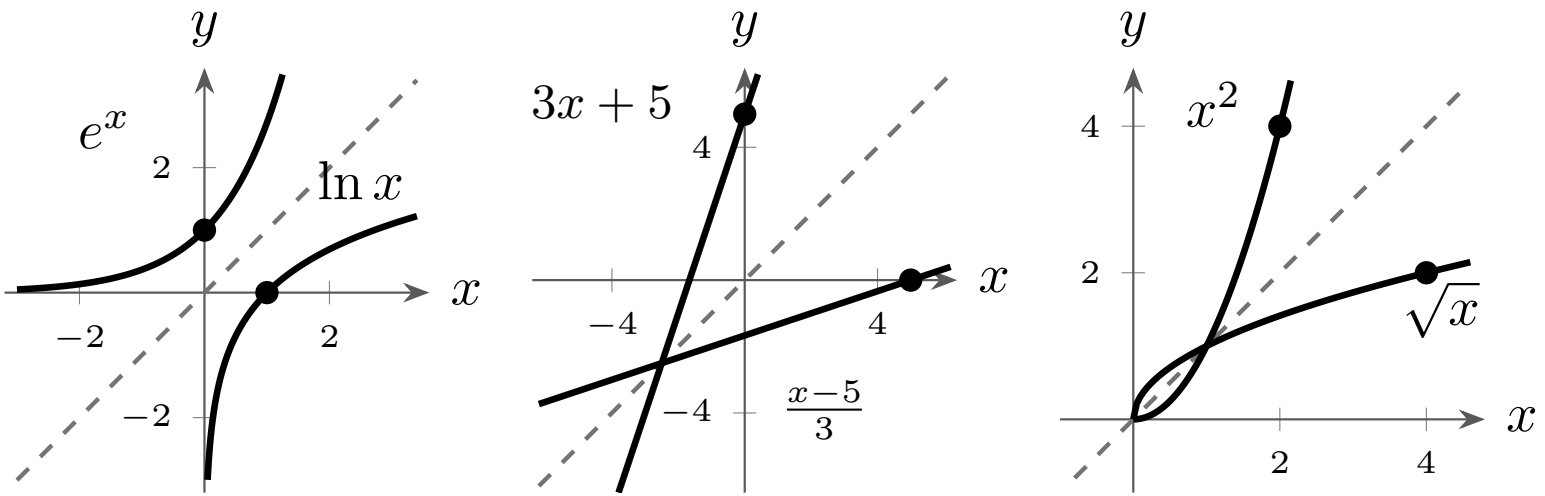
Function	**Domain** D	**Range** R
e^x	$\mathbb{R}$	$(0, \infty)$
$\ln x$	$(0, \infty)$	$\mathbb{R}$

This is the cleanest way to see why $\ln$ refuses non-positive inputs: e^x never *produces* a non-positive value, so $\ln$ is never *asked* about one.

Same story for $\sqrt{\cdot}$ as the inverse of x^2 restricted to $x \geq 0$: the restricted square has $R = [0, \infty)$, and that is exactly the square root's D.

An Inverse Is a Reflection Across $y = x$

Swapping input and output swaps every point: (a, b) on f becomes (b, a) on f^{-1}, a **mirror across** $y = x$ (dashed).



Each pair of dots is a point and its mirror image: $(0, 1)$ and $(1, 0)$, $(0, 5)$ and $(5, 0)$, $(2, 4)$ and $(4, 2)$.

The last panel draws x^2 only for $x \geq 0$, reflecting the **restricted** half is what produces $\sqrt{x}$.

Exercise: Inverses

The functions

$$f(x) = 4x - 7, \qquad g(x) = x^3 + 1, \qquad h(x) = x^2 - 4$$

Try it

1. Find f^{-1}, and check it.
2. Find g^{-1}.
3. Does h have an inverse on all of $\mathbb{R}$? If not, restrict its domain so that it does, and give the inverse there.
4. Solve $2^x = 32$ for x. (Which function undoes an exponential?)

Exercise: Solution

1 and 2

$y = 4x - 7 \Rightarrow x = \dfrac{y+7}{4}$, so $f^{-1}(x) = \dfrac{x+7}{4}$. Check:

$f^{-1}(4x-7) = \dfrac{4x-7+7}{4} = x$. ✓

$y = x^3 + 1 \Rightarrow x = \sqrt[3]{y-1}$, so $g^{-1}(x) = \sqrt[3]{x-1}$. (Cubing *is* one-to-one, so no restriction is needed.)

3. The one that needs a restriction

No: $h(-3) = h(3) = 5$, so h fails the horizontal line test. Restrict to $x \geq 0$; there h is one-to-one, and $y = x^2 - 4 \Rightarrow x = \sqrt{y+4}$, giving $h^{-1}(x) = \sqrt{x+4}$ with domain $[-4, \infty)$, which is precisely the range of the restricted h.

4. Undoing an exponential

Apply the logarithm base 2: $x = \log_2(32) = 5$, since $2^5 = 32$. Whenever the unknown sits in an **exponent**, the log is the tool that brings it down.

Functions of Several Variables

One Input Is Rarely Enough

Almost nothing worth predicting depends on a single number. A risk score depends on age *and* blood pressure *and* weight; a dose depends on weight *and* kidney function.

Function of p variables

$$f(x_1, x_2, \ldots, x_p) \;=\; \text{one output}$$

The rule takes an **ordered list** of p numbers and returns a single value. Everything in this lecture so far has been the case $p = 1$.

Example: body mass index

$$\mathrm{BMI}(w, h) = \frac{w}{h^2}$$

Two inputs (weight in kg, height in m), one output. The input is a **pair**, and the order matters, $\mathrm{BMI}(70, 1.8)$ is not $\mathrm{BMI}(1.8, 70)$.

"Exactly one output per input" is unchanged, what changed is that one input is now a whole list.

Notebook §1, evaluating a two-input rule

The Letters Are Ours to Choose

These three lines all say the same thing, **two inputs and one output**:

$$y = f(x_1, x_2), \qquad z = f(x, y), \qquad \text{BMI} = f(w, h)$$

Two conventions, both standard

Subscripts, $f(x_1, x_2, \ldots, x_p)$, scale to any number of inputs, so this is what data science uses once p is large.

Separate letters, $f(x, y)$ or $f(w, h)$, are handy when p is 2 or 3, because each input gets its own name and its own axis in a picture.

Watch out: a letter does not carry a job

In $y = f(x_1, x_2)$ the letter y is the **output**, plotted upward. In $z = f(x, y)$ that same letter y is an **input**, plotted along a horizontal axis, and z is the output.

Position tells you the role, not the letter: whatever sits inside the parentheses is an input, and whatever f equals is the output.

Notebook §2, the same function, three ways

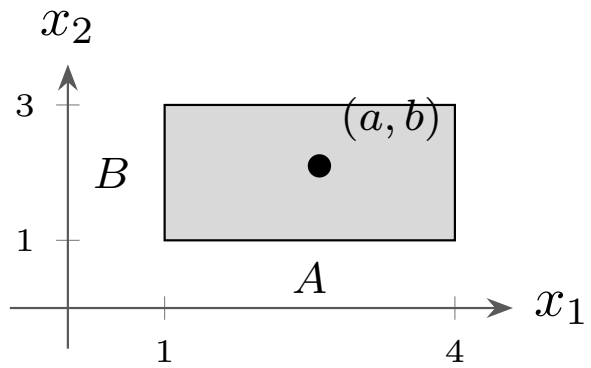
Where the Inputs Live: the Cartesian Product

Unit 1-2 built the set of all ordered pairs drawn from two sets, exactly where a two-variable function's inputs live.

Cartesian product

$$A \times B \;=\; \{\,(a,b) \;:\; a \in A,\; b \in B\,\}$$

and, for p sets, $A_1 \times A_2 \times \cdots \times A_p$ is the set of ordered p-tuples.



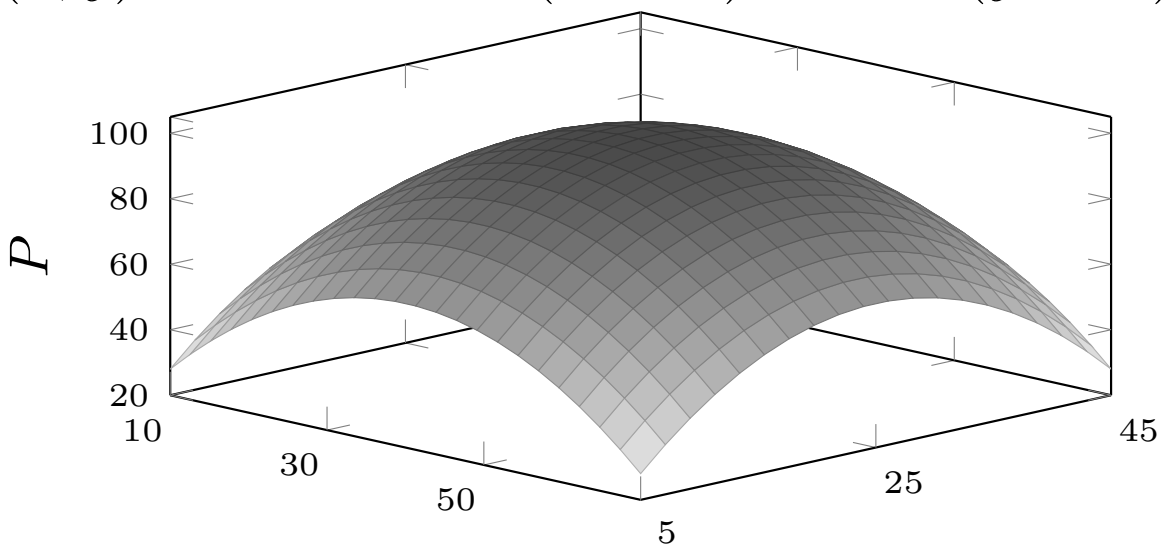
Here $A \times B$ is the shaded **rectangle**: every allowed x_1 paired with every allowed x_2. Taking $A = B = \mathbb{R}$ gives the plane $\mathbb{R}^2$, and p copies give $\mathbb{R}^p$.

Notebook §3, building a product set

The Graph of $f(x, y)$ Is a Surface

Two drugs are given together. Let x be the dose of drug A in mg, y the dose of drug B in mg, and $P(x, y)$ the percent of symptoms relieved. Too little of either does nothing, too much brings side effects that undo the benefit, so a simple model of the response is

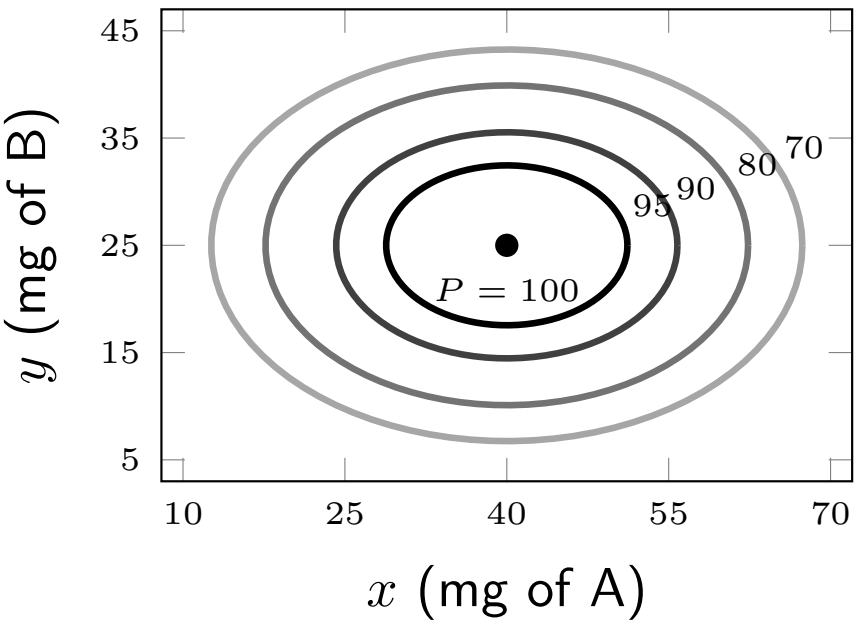
$$P(x, y) \;=\; 100 - 0.04\,(x - 40)^2 - 0.09\,(y - 25)^2.$$



A dose pair (x, y) is a **point on the floor**, and the **height of the sheet** above it is that pair's response. Both squares are subtracted, so the sheet is a **dome**.

Notebook §4, drag to rotate the dome

Level Curves: the Surface Seen from Above



A **level curve** collects every input pair with the *same* output:

$$\{(x, y) \ : \ P(x, y) = c\}.$$

Here each level curve is an **ellipse**, and they shrink toward a single point as c climbs. That point is the **maximum**, the best dose combination.

This is how a contour map shows a mountain: rings closing in on the summit. Crowded rings mean a steep climb, rings far apart a flat plateau where a dosing error costs little.

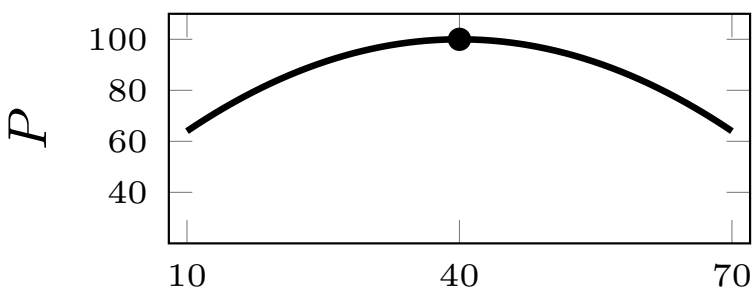
Reading that peak off the picture is what **optimization** does with algebra, in Unit 3.

Notebook §5, contour maps, and finding the peak

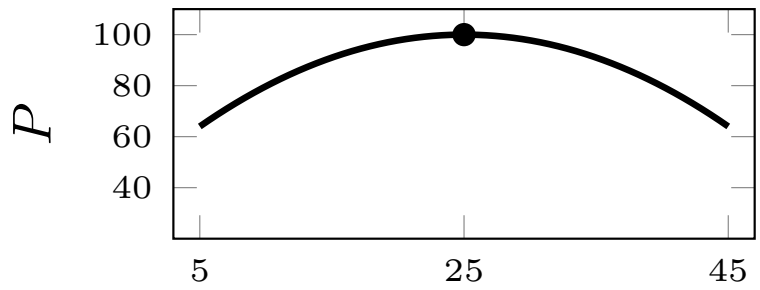
Slices: Hold One Input Fixed

Fix one variable at a number and a two-variable function collapses to an ordinary one-variable function, one of the families from earlier.

$y = 25$ fixed



$x = 40$ fixed



Both slices are parabolas, and both peak at 100

Holding B at 25 leaves $P = 100 - 0.04(x - 40)^2$, vertex at $x = 40$; holding A at 40 leaves $P = 100 - 0.09(y - 25)^2$, vertex at $y = 25$. A slice is a vertical cut through the dome, and clinically it asks **"I have fixed one drug, how much of the other?"**

Notebook §6, slide the slice yourself

Exercise: Reading a Surface

Try it

1. Why is $P(40, 25) = 100$ the largest value P can take? Argue from the formula, not the picture.
2. Compute $P(40, 35)$ and $P(50, 25)$. Which 10 mg overshoot costs the patient more, and why?
3. For $f(x, y) = x + y$, describe the surface and its level curves. Does it have a highest point?
4. For $g(x, y) = x^2 + y^2$, what shape are the level curves, and is the point at the center a maximum or a minimum?

Exercise: Solution

1 and 2: the peak, and the cost of a wrong dose

A square is never negative, so $0.04(x-40)^2 \geq 0$ and $0.09(y-25)^2 \geq 0$, and both are *subtracted* from 100. Any dose pair other than $(40, 25)$ subtracts something, so $P \leq 100$ always, with equality only at $x = 40$ and $y = 25$.

$P(40, 35) = 100 - 0 - 0.09(10)^2 = 91$, while
$P(50, 25) = 100 - 0.04(10)^2 - 0 = 96$. The same 10 mg overshoot costs more on **drug B**, because its coefficient 0.09 is larger: the response falls off faster in the y direction, which is exactly why the ellipses are shorter that way.

3 and 4: a plane and a bowl

$f(x, y) = x + y$ is a flat **plane**. Its level curves $x + y = c$ are **parallel lines** of slope -1, evenly spaced because the tilt never changes, and it has **no highest point**: keep increasing x and it keeps rising.

$x^2 + y^2 = c$ is a **circle** of radius $\sqrt{c}$, so the level curves are concentric rings and the surface is a **bowl**. Here the squares are *added*, so the center is a **minimum**, the dome flipped upside down.

Domain and Range, with Several Inputs

Nothing about D and R changes, except that D is now a set of **tuples**, carved out of a Cartesian product:

$$f : D \to R, \qquad D \subseteq A_1 \times A_2 \times \cdots \times A_p.$$

$\mathrm{BMI}(w, h) = w/h^2$

Weight and height are both positive, and h sits in a denominator, so $D = (0, \infty) \times (0, \infty)$, the open first quadrant. Every such pair gives a positive value, so $R = (0, \infty)$.

$g(x, y) = \sqrt{9 - x^2 - y^2}$

The root forces $9 - x^2 - y^2 \geq 0$, i.e. $x^2 + y^2 \leq 9$: D is the **disk** of radius 3, not a rectangle. Outputs run from 0 on the rim up to 3 at the center, so $R = [0, 3]$.

A domain in $\mathbb{R}^2$ is a **region**, not an interval, but still "the inputs the formula accepts."

Notebook §7, shading the allowed region

Exercise: Several Variables

Try it

1. $A = \{1, 2\}$ and $B = \{x, y, z\}$. List $A \times B$, and say how many elements it has.
2. Give D for $f(x, y) = \dfrac{1}{x - y}$, described in words.
3. Give D and R for $g(x, y) = \sqrt{x} + \ln y$.
4. A room's temperature depends on where you stand and when you measure. What are the inputs, and what is p?

Exercise: Solution

1 and 2

$A \times B = \{(1,x),(1,y),(1,z),(2,x),(2,y),(2,z)\}$, $2 \times 3 = 6$ elements. In general $|A \times B| = |A| \cdot |B|$.

f needs $x - y \neq 0$, so D is **the whole plane except the line** $y = x$: $D = \{(x,y) \in \mathbb{R}^2 : x \neq y\}$.

3. Two conditions, one for each coordinate

The root needs $x \geq 0$; the log needs $y > 0$. These act on separate coordinates, so the domain is a product: $D = [0,\infty) \times (0,\infty)$.

$\sqrt{x}$ covers $[0,\infty)$ and $\ln y$ covers all of $\mathbb{R}$, so their sum reaches every real value: $R = \mathbb{R}$.

4. The room

Three space coordinates and one time coordinate, $T(x,y,z,t)$, so $p = 4$, with $D \subseteq \mathbb{R}^4$ and one temperature out.

Functions in Machine Learning

What Machine Learning Is

Ordinary programming: **you** write the rule. But nobody can write the rule that turns a patient's chart into next year's glucose.

Machine learning

Machine learning is choosing a function from data. You supply a *family* of candidate rules and a pile of solved examples; the computer picks the member that best reproduces them, and you use it on unsolved cases.

The rule being chosen is a **function**, mapping **features** to a **label**:

$$\underbrace{y}_{\text{label}} = \underbrace{f}_{\text{model}}\Big(\underbrace{x_1, x_2, \ldots, x_p}_{\text{features}}\Big)$$

- A **feature** is a measured quantity used as an **input**: age, BMI, a lab value, a pixel.
- The **label** is the one quantity you want **out**. Learning from examples that already carry it is **supervised** learning.
- Which column plays which role is a **decision**, and a feature must exist **when you predict**: one measured afterward is a **leak**.
- "Learning" is not understanding, it is **minimization**.

Start with a Table

Six patients, three features each, and one label: their fasting glucose six months from now.

	features, the inputs x			**label**, the output y
patient	**age**	**BMI**	**systolic BP**	**glucose in 6 mo.**
A	54	27.4	138	110
B	61	22.8	122	101
C	47	31.5	145	117
D	58	24.1	118	102
E	43	33.2	150	116
F	66	28.6	131	116

- Three features here, so $p = 3$, and six patients, so $n = 6$.
- Each **row** is one patient: three numbers in, one number out.
- The double line is the whole problem. Everything left of it we have on the day we predict; the column right of it we will not see for six months.

A model is a rule that turns the left of that table into the right, for patients not in the table.

That Table Describes a Function

Read one row of the table as a single input and a single output:

$$f(\underbrace{54}_{\text{age}}, \underbrace{27.4}_{\text{BMI}}, \underbrace{138}_{\text{BP}}) \approx \underbrace{110}_{\text{glucose}}$$

- Three inputs, one output, so f is a function of $p = 3$ variables, the previous section's subject.
- Its **domain** is the set of measurement triples that make sense, the ages, BMIs and pressures a real patient could have, a region inside $\mathbb{R}^3$.
- Its **range** says what kind of answer comes out: a number here (**regression**); a label from a finite set such as $\{\text{diabetic}, \text{not diabetic}\}$ would make it **classification**.

The goal

Find one rule f that comes close on *all six* rows at once and, far more importantly, on patients we have never measured.

Notebook §9, features in, label out

Naming the Rows and Columns

Six rows is small enough to point at. A real table has thousands, so we need names, and each name carries **two** indices: which row, and which column.

	feature 1	**feature 2**	**feature 3**	**label**
patient 1	x_{11}	x_{12}	x_{13}	y_1
patient 2	x_{21}	x_{22}	x_{23}	y_2
$\vdots$				$\vdots$
patient n	x_{n1}	x_{n2}	x_{n3}	y_n

- x_{ij} is one **cell**, row i and column j. In the table two slides back, $x_{12} = 27.4$, patient A's BMI.
- A whole **row** is the tuple $x_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$, one input to f.
- y_i is that row's label, so n rows give the pairs $(x_1, y_1), \ldots, (x_n, y_n)$, the **training data**.

Read (x_i, y_i) out loud as "patient i's measurements, and patient i's answer." The goal is $f(x_i) \approx y_i$ for every i.

Inputs vs. Parameters

Look again at the linear function, now with its two roles labelled:

$$f(x) = \underbrace{m}_{\text{parameter}} \underbrace{x}_{\text{input}} + \underbrace{b}_{\text{parameter}}$$

- The **input** x changes from patient to patient.
- The **parameters** m and b are fixed once the model is trained, they identify *which* line, out of the whole family, we are using.

Two decisions, not one

Choosing a family (linear? exponential? something with a bend?) is modeling. **Choosing the parameters within that family** is fitting. The families earlier in this lecture are the menu you are choosing from.

Error Is a Function Too

Write the model's answer for row i as $\hat{y}_i$, read "y hat", to keep it apart from the truth y_i. The error is then a function of the *parameters*, not of x:

$$\hat{y}_i = mx_i + b \qquad \text{the prediction for row } i$$

$$E(m,b) = \sum_{i=1}^{n} \Big(\underbrace{y_i}_{\text{truth}} - \underbrace{\hat{y}_i}_{\text{prediction}} \Big)^2 \qquad \text{total squared miss}$$

$$= \sum_{i=1}^{n} \big(y_i - (mx_i + b)\big)^2 \qquad \text{the same thing, in full}$$

- In the last line every x_i and y_i is a fixed *number*; the only symbols still free are m and b.
- Squaring makes every miss count as positive, so errors cannot cancel out.
- **Training** is finding the m and b that make E smallest, and "find the input that minimizes a function" is **optimization**, the subject of Unit 3.
- Notice the reversal: the symbols that were *unknowns to estimate* a slide ago are the *inputs* here. That is the central move in all of model fitting.

Notebook §10, the error bowl and its minimum

Many Features, Same Function

With p features the linear model just grows a term per feature:

$$f(x_1, x_2, \ldots, x_p) = w_1 x_1 + w_2 x_2 + \cdots + w_p x_p + b$$

- The **inputs** are the x_j, one tuple from D, one row of the table.
- The **parameters** are now $w_1, \ldots, w_p$ and b: $p + 1$ numbers to fit instead of two, and the error $E(w_1, \ldots, w_p, b)$ is a function of all of them.
- Nothing conceptual changed: one output per input, a domain, a range, composition, and sometimes an inverse.

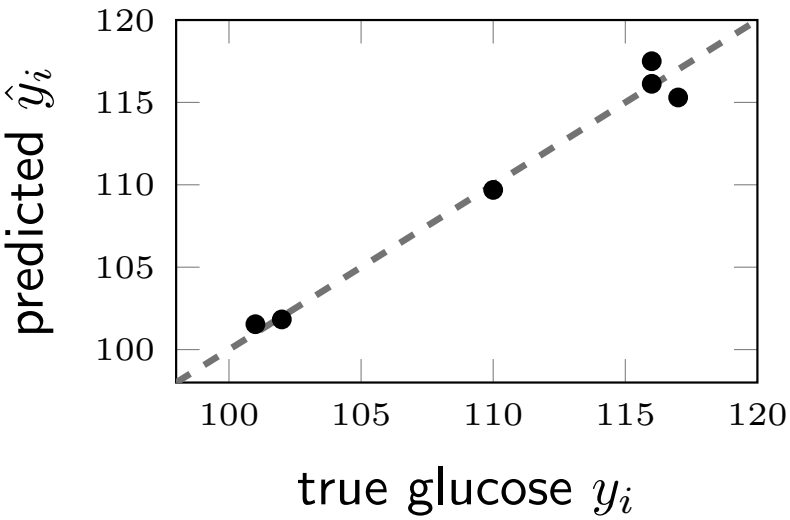
The bookkeeping is the only thing that grew, which is exactly what Unit 2's **vectors** and **matrices** are for, and why minimizing E over $p + 1$ parameters is Unit 3's problem rather than this lecture's.

Solving It: One Rule for All Six Patients

Minimizing E over the six rows of our table returns one specific member of the linear family:

$$\hat{y} \;=\; 0.458\,x_1 \;+\; 1.841\,x_2 \;+\; 0.181\,x_3 \;+\; 9.57$$

(x_1 age, x_2 BMI, x_3 systolic BP)



Each dot is one patient, true label across and prediction up. The dashed line is $\hat{y}_i = y_i$, **perfect prediction**, so a dot's gap from it is that patient's error.

How wrong is it?

$$\mathrm{SSE} = \textstyle\sum_i (y_i - \hat{y}_i)^2 = 5.57$$

$$\mathrm{RMSE} = \sqrt{\mathrm{SSE}/n} = 0.96 \text{ mg/dL}$$

$$\mathrm{MAE} = \tfrac{1}{n} \textstyle\sum_i |y_i - \hat{y}_i| = 0.72$$

RMSE is in the label's own units: a typical miss of about 1 mg/dL. But six rows against four parameters is nearly an even match, so this fit flatters itself.

Notebook §11, fit it yourself, and plot these residuals

Review

Review

What we covered: one variable

1. A **relation** is any set of ordered pairs; a **function** pairs *each* input with *exactly one* output.
2. Many-to-one is allowed, one-to-many is not, the **vertical line test**.
3. **Domain** D = valid inputs, **range** R = the outputs that actually occur. Watch $1/x$, $\sqrt{x}$, $\log x$.
4. Four **families**, polynomial, exponential, logarithmic, trigonometric, plus piecewise rules like ReLU, and **transformations**.
5. **Composition** $(f \circ g)(x) = f(g(x))$ runs inside out, and order matters; a pipeline is a composition.

Review

What we covered: quantifiers, inverses, several variables

6. **Quantifiers**: $\forall$ (every), $\exists$ (some), $\exists!$ (exactly one); negating one flips it, and order matters.
7. An **inverse** f^{-1} undoes f; it exists exactly when f is **one-to-one**, and its graph is the reflection across $y = x$.
8. With **several variables** the input is a tuple and D lives in a **Cartesian product**, one row of a data table is one such tuple.
9. Two inputs graph as a **surface**; **level curves** show it from above, and a **slice** at a fixed input is a one-variable curve.
10. A **model** is a function from a table's features to its label; its **error** is a function of the parameters, and minimizing it is Unit 3.

Notebooks: `U1-3_Functions-1_DomainRangeAndFamilies`, `-2_Composition`, `-3_Inverse`, `-4_FormulaExplorer`, `-5_SeveralVariables`. **Previous:** Unit 1-2, Sets. **Next:** Unit 1-4, **Series**.